Assignment of Waves and Oscillation

Date of Submission 9/4/2020

N.B.- Solve the problems correctly

1. An object undergoing simple harmonic motion takes 0.25 s to travel from one point of zero velocity to the next such point. The distance between those points is 36 cm. Calculate the (a) period, (b) frequency, and (c) amplitude of the motion.
2. The function x = (6.0 m) cos[(3 π rad/s)t + π /3 rad] gives the simple harmonic motion of a body. At t = 2.0 s, what are the (a) displacement, (b) velocity, (c) acceleration, and (d) phase of the motion? Also, what are the (e) frequency and (f) period of the motion?
3. An oscillator consists of a block of mass 0.500 kg connected to a spring. When set into oscillation with amplitude 35.0 cm, the oscillator repeats its motion every 0.500 s. find the (a) period, (b) frequency, (c) angular frequency, (d) spring constant, (e) maximum speed, and (f) magnitude of the maximum force on the block from the spring.
4. A simple harmonic oscillator consists of a block of mass 2.00 kg attached to a spring of spring constant 100 N/m. When t = 1.00 s, the position and velocity of the block are x = 0.129 m and v = 3.415 m/s.(a) What is the amplitude of the oscillations? What were the (b) position and (c) velocity of the block at t = 0 s?
5. An oscillator consists of a block attached to a spring (K = 400 N/m). At some time t, the positon (measured from the system's equilibrium location) velocity and acceleration of the block are x = 0.129 m and v = -13.6 m/s and a= -123 m/s^2. Calculate (a) frequency of the oscillations? (b) the mass of the block (c) amplitude of the motion?
6. When a body of unknown mass is attached to an ideal spring with force constant 120 N/m, it is found to vibrate with a frequency of 6.00 Hz. Find (a) the period of the motion; (b) the angular frequency; (c) the mass of the body.
7. A 2.00-kg, frictionless block is attached to an ideal spring with force constant 300 N/m. At t = 0 the spring is neither stretched nor compressed and the block is moving in the negative direction at 12.0 m/s. Find (a) the amplitude and (b) the phase angle. (c) Write an equation for the position as a function of time.

8. If a wave y(x, t) = (6.0 mm) sin(kx + (600 rad/s)t + φ) travels along a string, how much time does any given point on the string take to move between displacements y =+ 2.0 mm and y = - 2.0 mm?
9. A wave has an angular frequency of 110 rad/s and a wavelength of 1.80 m. Calculate (a) the angular wave number and (b) the speed of the wave.

10. A sinusoidal wave travels along a string. The time for a particular point to move from maximum displacement to zero is 0.170 s. What are (a) the period and (b) frequency? (c) The wavelength is 1.40 m; what is the wave speed?
11. A transverse sinusoidal wave is moving along a string in the positive direction of an x axis with a speed of 80 m/s. At t = 0, the string particle at x = 0 has a transverse displacement of 4.0 cm from its equilibrium position and is not moving. The maximum transverse speed of the string particle at x = 0 is 16 m/s. (a) What is the frequency of the wave? (b) What is the wavelength of the wave? If y(x, t) = y_m sin(kx ± ωt + φ) is the form of the wave equation, what are (c) y_m, (d) k, (e) ω, (f) φ, and (g) the correct choice of sign in front of ω?
12. The equation of a transverse wave traveling along a very long string is y = 6.0 sin(0.020πx + 4.0πt), where x and y are expressed in centimeters and t is in seconds. Determine (a)the amplitude, (b) the wavelength, (c) the frequency, (d) the speed, (e) the direction of propagation of the wave, and (f) the maximum transverse speed of a particle in the string.(g) What is the transverse displacement at x = 3.5 cm when = 0.26 s?

13. A wave traveling along a string is described by y(x, t) = 0.00327 sin(72.1x - 2.72t), in which the numerical constants are in SI units (0.00327 m, 72.1 rad/m, and 2.72 rad/s). (i) What is the amplitude, the wavelength, period, frequency and the velocity of this wave? (ii) What is the displacement y of the string at x= 22.5 cm and t = 18.9 s? (iii) What is u, the transverse velocity of the same element of the string, at that time? (iv)What is the transverse acceleration a of the same element at that time?

14. A sinusoidal wave of frequency 500 Hz has a speed of 350 m/s. (a) How far apart are two points that differ in phase by $\pi/3$ rad? (b) What is the phase difference between two displacements at a certain point at times 1.00 ms apart?

15. A string along which waves can travel is 2.70 m long and has a mass of 260 g. The wave speed 19.33 m/s. What must be the frequency of traveling waves of amplitude 7.70 mm for the average power to be 85.0 W?

16. A string has linear density μ = 525 g/m. We send a sinusoidal wave with wave speed = 9.26 m/s, frequency f = 120 Hz and amplitude y_m = 8.5 mm along the string. At what average rate does the wave transport energy?

17. If a wave y(x, t) = (6.0 mm) sin(kx + (600 rad/s)t + φ) travels along a string, how much time does any given point on the string take to move between displacements y =+ 2.0 mm and y = - 2.0 mm?

18. A wave has an angular frequency of 110 rad/s and a wavelength of 1.80 m. Calculate (a) the angular wave number and (b) the speed of the wave.

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